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Soil Test Report

Farmer Name: Maurice Barasa

Farm: Maurice Farm

Agent Livingstone Kibet

Crop: Macadamia

Date: 21st July

Test Results:

Parameter	Unit	Result	Guide Low	Guide High	Low	Optimum	High
Nitrogen	%	0.15	0.20	0.40			
Phosphorous	mg/kg	5.76	20	40			
Potassium	mg/kg	176	100	250			
pH		6.46	5.5	7.0			
EC	us/cm	49	200	500			

Analysis

- The soil has an optimum pH for Macadamia cultivation.
- Your potassium and nitrogen levels are sufficient for supporting robust flower production and early fruit development.
- However, phosphorus levels are critically low. This is especially important during flowering, as P supports reproductive growth, pollen tube development, and early nut formation.
- The EC levels are lower than expected.

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Recommendations

Apply **Triple Superphosphate (TSP) fertilizer** at a rate of 80 grams per tree just before the onset of the rains. This is essential to boost phosphorus levels in the soil, which is critical for flower development and nut set in macadamia.

During the fruit formation stage, apply **Muriate of Potash (MOP)** at a rate of 100 grams per tree. This will support healthy nut filling, enhance oil content, and strengthen the shell structure of the developing macadamia nuts.

At the beginning of the long rains, incorporate one debe (approximately 20 kg) of **well-composted manure** per tree. This improves soil organic matter, enhances microbial activity, and acts as a slow-release nutrient buffer throughout the growing season.

After flowering and during early fruiting, apply 80 grams per tree of Calcium Ammonium Nitrate (CAN) or UREA. This provides the nitrogen needed to support foliage activity, which is crucial for the tree to sustain its fruit load.

Apply a foliar mix containing Boron (0.3%), Zinc (0.5%), and Manganese every 3–4 weeks during flowering to improve flower retention, pollen viability, and nut set.

Conduct a soil test at year 2 to determine the level of nutrients of your soil.

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Best Practices

1. Canopy Management and Light Penetration: Maintain an open, well-structured canopy to allow adequate sunlight penetration, especially into the interior branches where flowers and young nuts develop.

Conduct selective pruning after harvest to remove any overcrowded, dead, or crossing branches. An east–west row orientation also enhances light interception during the flowering and fruiting period.

2. Water Management: Consistent soil moisture is critical during flowering and nut development. If rainfall is inadequate or irregular, consider supplemental irrigation. Avoid water stress during nut set and filling, as it can cause flower abortion or nut drop.

3. Pest and Disease Monitoring: Regularly scout for pests such as macadamia nut borer, thrips, or stink bugs, which can damage flowers and developing nuts. Also watch for fungal infections, especially during humid periods. Apply targeted treatments only when thresholds are surpassed, and always follow integrated pest management (IPM) principles.

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Input	Rate	Timing	Purpose
TSP (Triple Superphosphate)	80 g/tree	Before onset of long rains (Feb–Mar) or (Aug-Sep)	Boost phosphorus for flowering and nut set
MOP (Muriate of Potash)	100 g/tree	Early nut development (Apr–May)	Enhance nut filling, shell hardening, and kernel quality
Manure (well-decomposed)	1 debe/tree	Early long rains (Feb–Mar) & annually	Improve soil organic matter and nutrient buffering capacity
CAN or UREA	80 g/tree	Post-flowering (Apr–May)	Maintain nitrogen levels for leaf retention and photosynthesis

Note:

The recommendations given are based on the test result and the crop to be grown.